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Inventor(s)	Allwine; Scott F. et al.

Systems and methods for measuring characteristics of cryogenic electronic devices

Abstract

This disclosure relates to systems and methods for measuring impedance characteristics of a cryogenic device under test (DUT). A channel select circuit can be configured in a first state to electrically isolate a channel output circuit from the cryogenic DUT and in a second state to electrically couple the channel output circuit to the cryogenic DUT, and at least one resistor can be positioned along a transmission path that couples a pattern generator circuit to a channel output circuit that includes the channel select circuit. A controller can be configured to cause respective test current signals to be provided along the transmission path when the channel select circuit is in respective first and second states to establish respective first and second voltages across the at least one resistor, determine first and second impedance characteristics of the transmission path for determining an impedance of the cryogenic DUT.

Inventors: **Allwine; Scott F. (Leesburg, VA), Bagga; Sunny (Corona, CA), Cadwell; Brian J. (Pasadena, MD), Goodwin; Shaun Mark (East New Market, MD)**

Applicant: **NORTHROP GRUMMAN SYSTEMS CORPORATION** (Falls Church, VA)

Family ID: **88977635**

Assignee: **NORTHROP GRUMMAN SYSTEMS CORPORATION (Falls Church, VA)**

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Primary Examiner: Nguyen; Judy

Assistant Examiner: Alejnikov, Jr.; Robert P

Attorney, Agent or Firm: Tarolli, Sundheim, Covell & Tummino LLP

Background/Summary

RELATED APPLICATIONS (1) This application claims priority from U.S. patent application Ser. No. 17/830,665, filed 2 Jun. 2022, which is incorporated herein in its entirety.

TECHNICAL FIELD
(1) This disclosure relates to cryogenic electronic device testing.
BACKGROUND
(2) Cryogenic electronics refers to an operation of electronic devices, circuits, and systems at cryogenic temperatures. Cryogenic electronics (also referred to as low-temperature electronics, or cold electronics) can be based on semiconductive devices, superconductive devices, or a combination of the two.
SUMMARY
(3) In an example, a system can include a channel select circuit that can be configured in a first state to electrically isolate a channel output circuit from a cryogenic device under test (DUT) and in a second state to electrically couple the channel output circuit to the cryogenic DUT, at least one

resistor positioned along a transmission path that couples a pattern generator circuit to the channel output circuit that includes the channel select circuit and a controller. The controller can be configured to cause respective test current signals to be provided by the pattern generator circuit along the transmission path when the channel select circuit is in respective first and second states to establish respective first and second voltages across the at least one resistor, determine first and second impedance characteristics of the transmission path based on the established respective first and second voltages and an amount of current being provided by respective test current signals, and determine an impedance of the cryogenic DUT based on the first and second determined impedance characteristics of the transmission path.

(4) In another example, a system can include a measuring circuit that can be configured to measure a first voltage established across a resistor along a transmission path based on a first current signal. The first current test signal can be provided along the transmission path when a channel output circuit is isolated from a cryogenic DUT. The transmission path can couple a pattern generator circuit for generating current signals to a channel output circuit. The measuring circuit can be configured to measure a second voltage established across the resistor based on a second current signal. The second current test signal can be provided along the transmission path when the channel output circuit is electrically coupled to cryogenic DUT. The measuring circuit can be configured to determine first and second average voltages based on the measurements of the first and second voltages, respectively. The system includes a controller that can be configured to determine first and second impedance characteristics of the transmission path based on respective first and second average voltages and an amount of current being provided by respective first and second current test signals, and determine an impedance of the cryogenic DUT based on the first and second determined impedance characteristics of the transmission path.

(5) In a further example, a method can include causing a channel output circuit to be electrically isolated from a cryogenic DUT, controlling a test pattern generator circuit to provide a first current test signal along a transmission path so that the first current test signal flows through at least one resistor located in the transmission path to establish a first voltage across the at least one resistor, receiving a first determined average voltage based on measurements of the first voltage across the at least one resistor, determining first impedance characteristics of the transmission path based on the determined first average voltage and an amount of current being provided by the first current test signal, causing the channel output circuit to be electrically coupled to the cryogenic DUT such that the cryogenic DUT is part of the transmission path in response to determining the first impedance characteristics of the transmission path, controlling the test pattern generator circuit to provide a second current test signal along the transmission path so that the second current test signal flows through the cryogenic DUT and the at least one resistor to establish a first voltage across the at least one resistor, receiving a second determined average voltage based on measurements of the second voltage across the at least one resistor, determining second impedance characteristics of the transmission path based on the second average voltage and an amount of current being provided by the second current test signal, and determining impedance characteristics of the cryogenic DUT based on the first and second impedance characteristic of the transmission path.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is an example of a cryogenic DUT testing system.

(2) FIG. **2** is an example of another cryogenic DUT testing system.

(3) FIG. **3** is an example of a pattern generator circuit.

(4) FIG. **4** is an example of a method for determining impedance characteristics of a cryogenic

DUT.

(5) FIG. **5** is an example of another method for determining impedance characteristics of a cryogenic DUT.

DETAILED DESCRIPTION

(6) For testing of cryogenic electronics, a cryogenic DUT is connected via transmission paths (e.g., wires, conductive traces, etc.) to a test pattern generator. To test or verify a performance or characteristics of the cryogenic DUT, test patterns are generated by the test pattern generator and provided via the transmission paths to the cryogenic DUT. Nonidealities and transmission mismatching, and noise can limit a performance of a cryogenic DUT testing system, such that it becomes difficult to accurately measure impedance characteristics of the cryogenic DUT.

(7) Examples are described for measuring impedance characteristics of a cryogenic DUT. The term “impedance” as used herein refers to a phase shift or difference in a voltage or current operating angle. By way of example, the cryogenic DUT can be a Josephson Junction (JJ) device, however, in other examples, the cryogenic DUT can be another type of cryogenic device. In the examples herein, a cryogenic DUT testing system is configured to determine the impedance characteristics of the DUT by reducing or eliminating sources of DUT impedance error such as resulting from non-idealities in transmission paths and impedance mismatches between the transmission paths.

(8) For example, to determine the impedance characteristics of the cryogenic DUT, a controller of the cryogenic DUT testing system can be configured to control a current test pattern generator circuit of the system to output a current test signal to a channel output circuit of the system. The controller can be configured to cause the channel output circuit to be electrically isolated from the cryogenic DUT. By electrically isolating the channel output circuit from the cryogenic DUT impedance characteristics of transmission paths of the system can be estimated or determined according to the examples herein. The transmission paths can be used to electrically couple the current test pattern generator circuit to the channel output circuit (or circuitry therein). The channel output circuit can include a resistor that can be positioned or located along or within a respective transmission path over which current test signals can be provided to the cryogenic DUT (e.g., when the channel output circuit is electrically coupled to the cryogenic DUT). In some instances, the resistor can be positioned along or within the respective transmission path between the current test pattern generator circuit and the channel output circuit.

(9) The cryogenic DUT testing system includes a voltage sensing circuit and a voltage calculator circuit. The voltage sensing circuit can sense (e.g., measure) a voltage across the resistor to provide a sensed voltage signal. The voltage calculator circuit can be configured to sample the sensed voltage signal based on a sampling rate, which can be set by the controller. The voltage calculator circuit can be configured to determine an average voltage based on the samples of the sensed voltage signal. The voltage calculator circuit can be configured to output an average voltage signal indicative of the average voltage of the sensed voltage signal.

(10) The controller can be configured to determine transmission path characteristics (e.g., an impedance) for the transmission paths based on the average voltage signal and an amount of current being outputted by the current test pattern generator circuit. Because a resistance of the resistor and a voltage across the resistor is known, and the amount of current being outputted by the current test pattern generator circuit is also known, the controller can be configured to determine first impedance characteristics of the transmission paths (e.g., while the cryogenic DUT is electrically isolated from the channel output circuit and thus the transmission paths). The first impedance characteristics of the transmission paths can be representative of alternating current (AC) or direct current (DC) impedance characteristics of the transmission paths based on whether the current test pattern generator circuit outputs an AC or DC current test signal.

(11) In some examples, the controller can be configured to cause the channel output circuit to be in electrical connection with the cryogenic DUT. The impedance characteristics of transmission paths with the DUT in electrical connection (e.g., not electrically isolated from the channel output circuit

and thus the transmission paths) can be calculated in a same or similar manner as described herein for calculating the first impedance characteristics of the transmission paths. Thus, the controller can be configured to determine second impedance characteristics of the transmission paths during which the cryogenic DUT is electrically connected to the channel output circuit. The controller can be configured to determine an impedance difference based on the first and second impedance characteristics of the transmission paths. The impedance difference can be representative of impedance characteristics (e.g., an impedance, such as an AC or DC impedance) of the cryogenic DUT.

(12) Accordingly, the cryogenic DUT testing system described herein can accurately measure impedance characteristics of the cryogenic DUT (e.g., such as AC and/or DC impedance characteristics) by compensating for transmission path nonidealities and impedance mismatches between the transmission paths.

(13) FIG. **1** is an example of a cryogenic DUT testing system **100**. The system **100** includes a controller **102** and a current test pattern generator circuit **104**. The controller **102** can include a processor **106** (e.g., a central processing unit (CPU)) and memory **108**. The memory **108** can represent a non-transitory machine-readable memory (or other medium) that can be accessed by the processor **106** to execute impedance test logic **110** for testing impedance characteristics (e.g., AC and/or DC impedance characteristics) of a cryogenic DUT **112**. During impedance characteristic testing of the cryogenic DUT **112**, the processor **106** can execute an impedance characteristic calculator **114** that can be programmed to determine the impedance characteristics of the cryogenic DUT **112** based on different impedance characteristics of the transmission paths as described herein.

(14) The current test pattern generator circuit **104** can generate one or more different test patterns. For example, the current test pattern generator circuit **104** can generate a differential current test signal **116** that can correspond to a respective test pattern or be part of the respective test pattern. Thus, the current test pattern generator circuit **104** in some instances can be implemented as a differential current source. In additional or alternative examples, the current test pattern generator circuit **104** can be implemented as a bipolar current source such that current can be sinked or sourced with respect to the cryogenic DUT **112**. The current test pattern generator circuit **104** has a first output **118** and a second output **120**. In some instances, the first output **118** provides a current test signal **122** and the second output **120** receives the current test signal **122**. Thus, the first output **118** can output the current test signal **122** and the second output **120** can pull the current test signal **122**. Because the system **100** is for testing cryogenic DUTs, the current test signal **122** can be referred to as a high-precision test current. A “high-precision” test current is a current signal in a relatively low current range, such as a micro-amperes or nano-amperes range. In some instances, the current test signal **122** is an AC current test signal and in other instances, the current test signal **122** is a DC current test signal. The AC and DC current test signals can be used to determine AC and DC impedance characteristics of transmission paths and the cryogenic DUT **112** as described herein.

(15) The system **100** can further include a channel output circuit **124**. The channel output circuit **124** is coupled between the current test pattern generator circuit **104** and the cryogenic DUT **112**. The system **100** can include any number of channel output circuits similar to the channel output circuit **124**, as shown in FIG. **1**. A number of channel output circuits can be based on a number of cryogenic DUTs that are to be simultaneously or sequentially tested by the system **100**. Thus, each channel output circuit can be coupled to a respective current test pattern generator circuit for receiving a corresponding test pattern (e.g., one or more differential current test signals). In other examples, the current test pattern generator circuit **104** can be configured to provide a respective test pattern to each channel output circuit. Each channel output circuit can be coupled to a respective measuring circuit, such as a measuring circuit **126**, as shown in FIG. **1**.

(16) In the example of FIG. **1**, a single channel is shown for providing test patterns for testing a

single cryogenic DUT. While the example of FIG. **1** shows a single channel, in other examples, any number of channels (e.g., twelve (12) channels) can be used for testing a corresponding number of cryogenic DUTs. The cryogenic DUTs, and thus the cryogenic DUT **112** can be placed in a container, cooler, fridge, or chamber that can be configured to support a cryogenic environment. Each channel can be enabled or activated during cryogenic testing of the corresponding cryogenic DUT to provide a respective test pattern to the corresponding cryogenic DUT.

(17) By way of further example, the processor **106** can execute the impedance test logic **110** to provide a reference signal **128**. The reference signal **128** can directly or indirectly indicate an amount of current that is to be outputted by the current test pattern generator circuit **104**. For example, the reference signal **128** can indicate a voltage reference and the current test pattern generator circuit can include voltage-to-current (VI) converter that can provide the current test signal **122** based on the reference signal **128**. The current test pattern generator circuit **104** can provide the differential current test signal **116**, which includes the current test signal **122**, based on the reference signal **128**. The system **100** includes a current sensing circuit **130** to measure an amount of current being outputted by the current test pattern generator circuit **104**. The current sensing circuit **130** can sense (e.g., sample) the current test signal **122** to provide a sensed current test signal **132**. In some examples, the current test pattern generator circuit **104** can communicate to the controller **102** the amount of current being outputted as the current test signal **122**.

(18) The controller **102** can execute the impedance test logic **110** to provide a channel control signal **134**. The channel control signal **134** can control an electrical coupling of the channel output circuit **124** to the cryogenic DUT **112**. The channel control signal **134** can be provided to a channel select circuit **136** to electrically isolate the channel output circuit **124** from the cryogenic DUT **112**. By electrically isolating the channel output circuit **124** from the cryogenic DUT **112** impedance characteristics of transmission paths can be estimated or determined according to the examples described herein.

(19) In some instances, the channel select circuit **136** is configured in a first state (e.g., DUT isolating state) in which the channel output circuit **124** is electrically isolated from the cryogenic DUT **112**. In examples wherein the channel select circuit **136** is in the first state, the controller **102** can provide the channel control signal **134** to switch the channel select circuit **124** to a second state, thereby electrically coupling the channel output circuit **124** to the cryogenic DUT **112**. In some instances, the channel select circuit **136** is implemented as a single pole double throw relay (SPDT).

(20) As described herein, by electrically isolating the channel output circuit **124** from the cryogenic DUT **112** impedance characteristics of the transmission paths can be estimated or determined. The term “transmission path” as used herein can refer to a signal path (e.g., a wire, a trace, intermediate circuitry, and/or a combination thereof) along which a test signal or pattern can be provided. In the example of FIG. **1**, a first transmission path **138** can extend from the first output **118** of the current test pattern generator circuit **104** to a first input **140** of the channel select circuit **136**. A second transmission path **142** can extend from the second output **120** of the current test pattern generator circuit **104** through a resistor **144** to a second input **146** of the channel select circuit **136**, as shown in FIG. **1**. In some instances, the first and second inputs **140** and **146** correspond to the inputs of the channel output circuit **124** and in these examples, the resistor **144** can be located between the current test pattern generator circuit **104** and the channel output circuit **124**. Each channel output circuit can include a resistor that is positive along a respective transmission path that runs therein. Each resistor can be positioned along a respective transmission path so that the resistor is in series with an output of a respective current test pattern generator circuit and thus in series with a respective cryogenic DUT (e.g., when electrically coupled).

(21) By way of example, the channel select circuit **136** can include circuitry that can couple the first and second inputs **140** and **146** internally so that the first and second inputs **140** and **146** are shorted therein. In some instances, the first and second inputs **140** and **146** of the channel select

circuit **136** are shorted in response to the channel control signal **134**. For example, the channel select circuit **136** can include a switch that can be positioned between the first and second inputs **140** and **146**. The switch can be actuated and thus closed in response to the channel control signal **134** to short the first and second inputs **140** and **146**. When the switch is in an open state, the first and second inputs **140** and **146** can be electrically coupled to respective first and second outputs of the channel select circuit **136** so that the cryogenic DUT **112** can be provided test patterns. By shorting the first and second inputs **140** and **146**, a closed loop circuit **148** is formed in which the current test signal **122** flows from the first output **118** of the current test pattern generator circuit **104** through the short formed between the first and second inputs **140** and **146** through the resistor **144** back to the second output **120** of the current test pattern generator circuit **104**. A voltage can be established across the resistor **144** based on the current test signal **122**.
(22) The measuring circuit **126** includes a voltage sensing circuit **150** that can sense (e.g., measure) the voltage across the resistor **144** to provide a sensed voltage signal **152**. The voltage sensing circuit **150** can provide the sensed voltage signal **152** with a sufficient gain that raises the sensed voltage signal **152** above a noise floor of a voltage calculator circuit **154**. This is because the sensed voltage signal **152** is in a micro-volt range or nano-volt range as the voltage across the resistor **144** is based on a current that is in a micro-amperes or nano-amperes range. The voltage calculator circuit **154** can be configured to sample the sensed voltage signal **152** based on a sampling rate, which can be set by the controller **102**. The voltage calculator circuit **154** can be configured to determine an average voltage based on the samples of the sensed voltage signal **152**. The voltage calculator circuit **154** can be configured to output an average voltage signal **156** indicative of an average voltage of the sensed voltage signal **152**. In some examples, the voltage calculator circuit **154** can be configured to output an RMS voltage based on the samples of the sensed voltage signal **152**. A polarity of the average voltage signal **156** can indicate in which direction a current flows through the resistor **144**. For example, if the current test signal **122** is provided from the second output **120** of the current test pattern generator circuit **104** to the resistor **144**, the average voltage signal **156** can have a negative polarity.
(23) The controller **102** can be configured to receive the average voltage signal **156**. The impedance test logic **110** can cause the controller **102** to determine transmission path characteristics based on the average voltage signal **156** and an amount of current being outputted by the current test pattern generator circuit **104**. The amount of current outputted by the current test pattern generator circuit **104** can be determined based on the sensed current test signal **132**. For example, the processor **106** can execute the impedance characteristic calculator **114** to determine the transmission path characteristics for the first and second transmission paths **138** and **142**.
(24) Because a resistance of the resistor **144** and a voltage across the resistor **144** is known (based on the average voltage signal **156**), and the amount of current outputted by the current test pattern generator circuit **104** is also known (based on the sensed current test signal **132**), the impedance characteristic calculator **114** can be programmed to determine first impedance characteristics representative of an overall impedance of the first and second transmission paths **138** and **142**. In examples wherein the current test signal **122** is the DC current test signal the first determined impedance characteristics can characterize an overall DC impedance of the first and second transmission paths **138** and **142**. In examples wherein the current test signal **122** is the AC current test signal the first determined impedance characteristics can characterize an overall AC impedance of the first and second transmission paths **138** and **142**.
(25) In some examples, the controller **102** can be configured to cause the channel output circuit **124** to be in electrical connection with the cryogenic DUT **112**. For example, the controller **102** can be configured to stop providing the channel control signal **134** to remove the short between the first and second inputs **140** and **146** of the channel select circuit **136**. The channel select circuit **136** in some instances can be configured to couple the first and second inputs **140** and **146** to respective outputs **158** and **160** of the channel select circuit **136** in response to not receiving the channel

control signal **134**. In other examples, the controller **102** can be configured to provide a different or another channel control signal **134** to cause the channel select circuit **136** to couple the first and second inputs **140** and **146** to respective outputs **158** and **160** of the channel select circuit **136** therein. Once the first and second inputs **140** and **146** are coupled to respective outputs **158** and **160** to which corresponding inputs of the cryogenic DUT **112** are coupled, the cryogenic DUT **112** can be electrically coupled to the first and second transmission paths **138** and **142**.

(26) Overall impedance characteristics for the first and second transmission paths **138** and **142** with the cryogenic DUT **112** in electrical connection (e.g., not electrically isolated) can be calculated in a same or similar manner as described herein for calculating the first impedance characteristics for the first and second transmission paths **138** and **142**. Thus, the controller **102** can be configured to determine second impedance characteristics representative of an overall impedance of the first and second transmission paths **138** and **142** during which the cryogenic DUT **112** is electrically connected to the channel output circuit **124**. In some instances, the AC and DC impedance characteristics for the first and second transmission paths **138** and **142** can be determined in a same or similar manner as described herein. Because the cryogenic DUT **112** is electrically coupled to the channel output circuit **124**, impedance characteristics of one of the first and second transmission paths **138** and **142** can change as the cryogenic DUT **112** can influence an impedance of at least one of the first and second transmission paths **138** and **142**.

(27) The impedance characteristic calculator **114** can be programmed to determine an impedance difference based on the first and second determined impedance characteristics for the first and second transmission paths **138** and **142**. The impedance difference can be representative of an impedance of the cryogenic DUT **112**. For example, the impedance characteristic calculator **114** can be programmed to subtract a second impedance from the first impedance to compute an impedance difference corresponding to the impedance of the cryogenic DUT **112**. In some examples, the impedance characteristic calculator **114** can be programmed to compute a DUT current noise contribution and DUT voltage drop. For example, the DUT voltage drop can be computed based on a difference in the voltage across the resistor **144** when the cryogenic DUT **112** is electrically isolated from the channel output circuit **124** and when the cryogenic DUT **112** is electrically coupled to the channel output circuit **124**. In additional or alternative examples, the controller **102** can be configured to determine the DUT current noise contribution by applying a Fast Fourier Transform (FFT) based on the average voltage signal **156**. The controller **102** can be configured to compute the DUT current noise contribution of the cryogenic DUT **112** based on a difference in the DUT current noise contribution when the cryogenic DUT **112** is electrically coupled to the channel output circuit **124** and electrically isolated from the channel output circuit **124**.

(28) Thus, the controller **102** can be configured to measure DUT electrical attributes, such as current (e.g., based on the resistance of the resistor **144** and the average voltage signal **156**), a phase (e.g., based on the current test signal **122** and the average voltage signal **156**, and power (e.g., consumed by the cryogenic DUT **112**, such as based on the current test signal **122** and the average voltage signal **156**). Because these attributes can change depending on a DUTs performance, proper operating points of the DUT can be determined. Moreover, the measured DUT electrical attributes can be used for diagnosing operational issues of the cryogenic DUT **112**.

(29) Accordingly, the cryogenic DUT testing system **100** can accurately measure impedance characteristics of the cryogenic DUT **112** (e.g., such as AC and/or DC impedance characteristics) by compensating for transmission path nonidealities and impedance mismatches between the first and second transmission paths **138** and **142**.

(30) FIG. **2** is an example of another cryogenic DUT testing system **200** that can be used for testing a performance and/or characteristics of a cryogenic DUT (e.g., the cryogenic DUT **112**, as shown in FIG. **1**). Thus, in some examples, reference can be made to FIG. **1** in the example of FIG. **2**. The system **200** can be implemented with a temperature compensation circuit **204** that can be

configured to compensate for temperature effects on a differential current test signal **206** provided to the cryogenic DUT during cryogenic DUT testing. The temperature compensation circuit **204** can correspond to a temperature compensation circuit as described in co-pending application entitled “Temperature Compensated Current Source for Cryogenic Electronic Testing” (application Ser. No. 17/830,697, now U.S. Pat. No. 11,789,065) and filed concurrently with this application, which is incorporated herein by reference in its entirety. The temperature compensation circuit **204** can be used to mitigate temperature drift effects on the differential current test signal **206** and provide the cryogenic DUT with stable testing currents.

(31) The system **200** includes a voltage source circuit **208**. The voltage source circuit **208** can be configured to provide a differential temperature-compensated voltage **210**, which can be used to provide a temperature compensated differential current test signal (e.g., in some instances corresponding to the differential current test signal **116**, as shown in FIG. **1**). In the example of FIG. **2**, the voltage source circuit **208** includes the temperature compensation circuit **204**, however, in other examples, the temperature compensation circuit **204** may be located outside the voltage source circuit **208**. For example, the temperature compensation circuit **204** can be implemented on a separate die and can be coupled to another die on which the voltage source circuit **208** can be implemented. The temperature compensation circuit **204** can provide a temperature compensation signal **212** to a differential voltage generator **214**.

(32) The differential voltage generator **214** can be configured to receive the temperature compensation signal **212** and a reference signal **216** that can provide a reference voltage. In some examples, the reference signal **216** can correspond to the reference signal **128**, as shown in FIG. **1**. The differential voltage generator **214** can be configured to output the differential temperature compensated voltage **210** based on the reference signal **216** and the temperature compensation signal **212**. In some examples, the reference signal **216** is a digital signal and the differential voltage generator **214** includes a digital to analog (DAC) converter. By way of example, the DAC converter can be a 20-bit DAC converter. In other examples, a DAC converter with a different resolution type can be used. The system **200** can include a main controller **218** that can be configured to provide the reference signal **216**. The main controller **218** can correspond to the controller **102**, as shown in FIG. **1**.

(33) The system **200** can further include a current output circuit **220**. The current output circuit **220** can include a VI converter **222**. In some examples, the differential voltage generator **214** and the VI converter **222** can form a current test pattern generator circuit **224**, such as the current test pattern generator circuit **104**, as shown in FIG. **1**. The VI converter **222** can be configured to provide the differential current test signal **206** based on the differential temperature compensated voltage **210**. The current output circuit **220** further includes a channel output circuit **226** that can be configured to provide the differential current test signal **206** as a temperature compensated differential test current to the cryogenic DUT.

(34) In the example of FIG. **2**, a single channel is shown for providing a differential test current for testing a single cryogenic DUT. Thus, while the example of FIG. **2** is described with respect to a single channel, in other examples, any number of channels (e.g., twelve (12) channels) can be used for testing a corresponding number of cryogenic DUTs. The cryogenic DUTs can be placed in a container, cooler, fridge, or chamber that is configured to support a cryogenic environment. Each channel can be enabled or activated during cryogenic testing of the corresponding cryogenic DUT by a channel controller **228** to provide a respective differential test current to the corresponding cryogenic DUT.

(35) By way of example, the current test pattern generator circuit **224** has a first output **230** and a second output **232**. In some instances, the first output **230** provides a current test signal **234** and the second output **232** receives the current test signal **122**. The current test signal **234** can be similar to the current test signal **122**, as shown in FIG. **1**. Thus, the first output **230** can output the current test signal **234** and the second output **232** can pull the current test signal **234**. Because the system **200** is

for testing cryogenic DUTs, the current test signal **234** can be referred to as a high-precision test current. In some instances, the current test signal **234** is an AC current test signal and in other instances is a DC current test signal. The AC and DC current test signals can be used to determine AC and DC impedance characteristics of transmission paths and the cryogenic DUT such as described herein.
(36) In further examples, the current output circuit **220** includes a differential amplifier **236** and an average voltage calculator circuit **238**. The differential amplifier **236** can correspond to the voltage sensing circuit **150**, as shown in FIG. **1**, and the average voltage calculator circuit **238** can correspond to the voltage calculator circuit **154**, as shown in FIG. **1**. In some instances, the differential amplifier **236** and the average voltage calculator circuit **238** can define a measuring circuit **240**, such as the measuring circuit **126**, as shown in FIG. **1**.
(37) Continuing with the example of FIG. **2**, the main controller **218** can be configured to execute impedance test logic (e.g., the impedance test logic **110**, as shown in FIG. **1**) for testing impedance characteristics of the cryogenic DUT to provide the reference signal **216**. The current test pattern generator circuit **224** can provide the differential current test signal **206**, which includes the current test signal **234**, based on the reference signal **216**. The system **200** includes a current sensing circuit **242** to measure an amount of current being outputted by the current test pattern generator circuit **224**. The current sensing circuit **242** can sense (e.g., sample) the current test signal **234** to provide a sensed current test signal **244**. The channel controller **228** can receive and communicate the sensed current test signal **244** to the main controller **218**.
(38) The main controller **218** can be configured to provide a channel control signal **246**. The channel control signal **246** can correspond to the channel control signal **134**, as shown in FIG. **1**. The main controller **218** can provide the channel control signal **246** to the channel controller **228**. The channel control signal **246** can control an electrical coupling of the channel output circuit **226** to the cryogenic DUT. The channel control signal **246** can be provided by the channel controller **228** to a channel select circuit **248** to electrically isolate the channel output circuit **226** from the cryogenic DUT. By electrically isolating the channel output circuit **226** from the cryogenic DUT impedance characteristics of transmission paths can be estimated or determined according to the examples described herein.
(39) In some instances, the channel select circuit **248** is configured in a first state (e.g., DUT isolating state) in which the channel output circuit **226** is electrically isolated from the cryogenic DUT. In examples wherein the channel select circuit **248** is in the first state, the channel controller **228** can provide the channel control signal **246** to switch the channel select circuit **248** to a second state, thereby electrically coupling the channel output circuit **124** to the cryogenic DUT. In some instances, the channel select circuit **248** includes a relay **250**, such as a SPDT relay.
(40) In the example of FIG. **2**, a first transmission path **252** can extend from the first output **230** of the current test pattern generator circuit **224** through a first resistor **254**) to a first input **256** of the relay **250**. A second transmission path **258** can extend from the second output **232** of the current test pattern generator circuit **224** through a second resistor **260** to a second input **262** of the relay **250**, as shown in FIG. **2**. The first and second resistors **254** and **260** can have respective resistances (labelled as “R**1**” and “R**2**” in the example of FIG. **2**). The first and second resistors **254** and **260** can have complementary resistances. In an example, the relay **250** is a SPDT relay.
(41) In some instances, the first and second inputs **256** and **262** of the relay **250** are shorted to form a short **263** in response to the channel control signal **246**. By shorting the first and second inputs **256** and **262**, the current test signal **234** flows from the first output **230** of the current test pattern generator circuit **224** through the first resistor **254** and the short **263** formed between the first and second inputs **256** and **262** through the second resistor **260** to the second output **232** of the current test pattern generator circuit **224**. A respective voltage can be established across the first and second resistors **254** and **262** based on the current test signal **234**. The differential amplifier **236** can be configured to sense (e.g., measure) the voltage across the first resistor **254** to provide a sensed

voltage signal **264**. While the example of FIG. **2** illustrates the voltage across the first resistor **254** being sensed in other examples the voltage across the second resistor **260** can be sensed to provide the sensed voltage signal **264**.

(42) The differential amplifier **236** can provide the sensed voltage signal **264** with a sufficient gain that raises the differential amplifier **236** above a noise floor of the average voltage calculator circuit **238**. This is because the sensed voltage signal **264** is in a micro-volt range or nano-volt range as the voltage across the first resistor **254** is based on a current that is in a micro-amperes or nano-amperes range. The average voltage calculator circuit **238** can be configured to sample the sensed voltage signal **264** based on a sampling rate, which can be set by the channel controller **228**. The average voltage calculator circuit **238** can be configured to determine an average voltage based on the samples of the sensed voltage signal **264**. The average voltage calculator circuit **238** can be configured to output an average voltage signal **266** indicative of an average voltage of the sensed voltage signal **264**. A polarity of the average voltage signal **266** can indicate in which direction current flows through the first resistor **254**. For example, if the current test signal **234** is provided from the second output **232** of the current test pattern generator circuit **224** and thus flows in an opposite direction if provided from the first output **230**, the average voltage signal **266** can have a negative polarity.

(43) The channel controller **228** can be configured to provide the average voltage signal **266** to the main controller **218**. The main controller **218** can be configured to determine transmission path characteristics based on the average voltage signal **266** and an amount of current outputted by the current test pattern generator circuit **224**. The amount of current outputted by the current test pattern generator circuit **224** can be determined based on the sensed current test signal **244**. For example, the main controller **218** can execute a characteristic calculator (e.g., the characteristic calculator **114**, as shown in FIG. **1**) which can be programmed to determine the transmission path characteristics for the first and second transmission paths **252** and **258**.

(44) Because a resistance of the first resistor **254** and a voltage across the first resistor **254** is known (based on the average voltage signal **266**), and the amount of current outputted by the current test pattern generator circuit **224** is also known (based on the sensed current test signal **244**), the main controller **218** can be configured to determine first impedance characteristics representative of an overall impedance of the first and second transmission paths **252** and **258**. In examples wherein the current test signal **234** is the DC current test signal the first determined impedance characteristics can characterize an overall DC impedance of the first and second transmission paths **252** and **258**. In examples wherein the current test signal **234** is the AC current test signal the first determined impedance characteristics can characterize an overall AC impedance of the first and second transmission paths **252** and **258**.

(45) In some examples, the main controller **218** can be configured to cause the channel output circuit **226** to be in electrical connection with the cryogenic DUT **112**. For example, the main controller **218** can communicate with the channel controller **228** to cause the channel controller **28** to stop providing the channel control signal **246** to remove the short **233** between the first and second inputs **256** and **262** of the relay **250**. Overall impedance characteristics for the first and second transmission paths **252** and **258** with the DUT in electrical connection (e.g., not electrically isolated) can be calculated in a same or similar manner as described herein for calculating the first impedance characteristics for the first and second transmission paths **252** and **258**. Thus, the main controller **218** can be configured to determine second impedance characteristics representative of an overall impedance of the first and second transmission paths **252** and **258** during which the cryogenic DUT is electrically connected to the channel output circuit **226**. In some instances, the AC and DC impedance characteristics for the first and second transmission paths **252** and **258** can be determined in a same or similar manner as described herein. Because the cryogenic DUT is electrically coupled to the channel output circuit **226**, impedance characteristics of one of the first and second transmission paths **252** and **258** can change as the cryogenic DUT can influence an

impedance of at least one of the first and second transmission paths **252** and **258**.
(46) The main controller **218** can be configured to determine an impedance difference based on the first and second determined impedance characteristics for the first and second transmission paths **252** and **258**. The impedance difference can be representative of an impedance of the cryogenic DUT. For example, the main controller **218** can be configured to subtract a second impedance from a first impedance to compute an impedance difference corresponding to the impedance of the cryogenic DUT.
(47) Accordingly, the cryogenic DUT testing system **200** can accurately measure impedance characteristics of the cryogenic DUT (e.g., such as AC and/or DC impedance characteristics) by compensating for transmission path nonidealities and impedance mismatches between the first and second transmission paths **252** and **258**.
(48) FIG. **3** is an example of a current test pattern generator circuit **300**. The current test pattern generator circuit **300** can be the current test pattern generator circuit **104**, as shown in FIG. **1**, or the current test pattern generator circuit **224**, as shown in FIG. **2**. Thus, in some examples, reference can be made to FIGS. **1-2** in the example of FIG. **3**. The current test pattern generator circuit **300** includes a single-ended-to-differential converter **302**, a DAC controller **304**, and a temperature compensation circuit **306**, which can correspond to the temperature compensation circuit **204**, as shown in FIG. **2**. The single-ended-to-differential converter **302** and the DAC controller **304** can define or form the differential voltage generator **214**, as shown in FIG. **2**. Thus, the DAC controller **304** can be configured to receive a reference signal **308**. The reference signal **308** can be representative of a digital signal and characterize a reference voltage to be provided by the DAC controller **304** to the single-ended-to-differential converter **302**. In some instances, the reference signal **308** corresponds to the reference signal **128**, as shown in FIG. **1**. The DAC controller **304** can be configured to provide an intermediate reference voltage **310** to the single-ended-to-differential converter **302** based on the reference signal **308**. The temperature compensation circuit **306** can correspond to the temperature compensation circuit **204**, as shown in FIG. **2**, and can be configured to provide a temperature compensation signal **312**. The temperature compensation signal **312** can be the temperature compensation signal **212**, as shown in FIG. **2**. The single-ended-to-differential converter **302** can be configured to output a differential temperature compensated voltage **314** based on a difference between the intermediate reference voltage **310** and the temperature compensation signal **312**. The differential temperature compensated voltage **314** can correspond to the differential temperature compensated voltage **210**, as shown in FIG. **2**.
(49) In view of the foregoing structural and functional features described above, example methods will be better appreciated with reference to FIGS. **4-5**. While, for purposes of simplicity of explanation, the example methods of FIGS. **4-5** are shown and described as executing serially, it is to be understood and appreciated that the present examples are not limited by the illustrated order, as some actions could in other examples occur in different orders, multiple times and/or concurrently from that shown and described herein. Moreover, it is not necessary that all described actions be performed to implement each method below.
(50) FIG. **4** is an example of a method **400** for determining impedance characteristics of a cryogenic DUT such as the cryogenic DUT **112**, as shown in FIG. **1**. Thus, in some examples, reference can be made to FIGS. **1-3** in the example of FIG. **4**. The method **400** can begin at **402** by electrically isolating a channel output circuit (e.g., the channel output circuit **124**, as shown in FIG. **1**) from the cryogenic DUT. At **404**, providing a first current test signal (e.g., the current test signal **122**, as shown in FIG. **1**, or the current test signal **234**, as shown in FIG. **2**) along a transmission path (e.g., the first and second transmission paths **138** and **142**, as shown in FIG. **1**) such that the first current test signal flows through at least one resistor (the resistor **144**, as shown in FIG. **1**) located in the transmission path.
(51) At **406**, measuring a first voltage across the at least one resistor based on the first current test signal. At **408**, determining a first average voltage based on measurements of the first voltage

across the resistor. At **410**, determining first impedance characteristics of the transmission path based on the first average voltage and an amount of current being provided by the first current test signal. At **412**, electrically coupling the channel output circuit to the cryogenic DUT such that the cryogenic DUT is part of the transmission path. At **414**, providing a second current test signal (e.g., the current test signal **122**, as shown in FIG. **1**, or the current test signal **234**, as shown in FIG. **2**) along the transmission path such that the second current test signal flows through the cryogenic DUT and the at least one resistor. At **416**, measuring a second voltage across the at least one resistor based on the second current test signal. At **418**, determining a second average voltage based on measurements of the second voltage across the at least one resistor. At **420**, determining second impedance characteristics of the transmission path based on the second average voltage and an amount of current being provided by the second current test signal. At **422**, determining impedance characteristics of the cryogenic DUT based on the determined first and second impedance characteristic of the transmission path.

(52) FIG. **5** is an example of another method **500** for determining impedance characteristics of a cryogenic DUT, such as the cryogenic DUT **112**, as shown in FIG. **1**. The method **500** can be implemented by a controller, such as the controller **102**, as shown in FIG. **1**, or the main controller **218**, as shown in FIG. **2**. Thus, in some examples, reference can be made to FIGS. **1-3** in the example of FIG. **5**. At **502**, causing a channel output circuit (e.g., the channel output circuit **124**, as shown in FIG. **1**) to be electrically isolated from the cryogenic DUT. At **504**, controlling a test pattern generator circuit (e.g., the current test pattern generator circuit **104**, as shown in FIG. **1**) to provide a first current test signal (e.g., the current test signal **122**, as shown in FIG. **1**, or the current test signal **234**, as shown in FIG. **2**) along a transmission path (e.g., the first and second transmission paths **138** and **142**, as shown in FIG. **1**) so that the first current test signal flows through at least one resistor (the resistor **144**, as shown in FIG. **1**) located in the transmission path to establish a first voltage across the at least one resistor. At **506**, receiving a first determined average voltage based on measurements of the first voltage across the at least one resistor. At **508**, determining first impedance characteristics of the transmission path based on the determined first average voltage and an amount of current being provided by the first current test signal.

(53) At **510**, causing the channel output circuit to be electrically coupled to the cryogenic DUT such that the cryogenic DUT is part of the transmission path. At **512**, controlling the test pattern generator circuit to provide a second current test signal along the transmission path so that the second current test signal flows through the cryogenic DUT and the at least one resistor to establish a first voltage across the at least one resistor. At **514**, receiving a second determined average voltage based on measurements of the second voltage across the at least one resistor. At **516**, determining second impedance characteristics of the transmission path based on the second average voltage and an amount of current being provided by the second current test signal. At **518**, determining impedance characteristics of the cryogenic DUT based on the first and second impedance characteristic of the transmission path.

(54) What has been described above are examples. It is, of course, not possible to describe every conceivable combination of components or methodologies, but one of ordinary skill in the art will recognize that many further combinations and permutations are possible. Accordingly, the disclosure is intended to embrace all such alterations, modifications, and variations that fall within the scope of this application, including the appended claims. As used herein, the term “includes” means includes but not limited to, the term “including” means including but not limited to. The term “based on” means based at least in part on. Additionally, where the disclosure or claims recite “a,” “an,” “a first,” or “another” element, or the equivalent thereof, it should be interpreted to include one or more than one such element, neither requiring nor excluding two or more such elements.

Claims

1. A system comprising: a channel select circuit configured in a first state to electrically isolate a channel output circuit from a cryogenic device under test (DUT) and in a second state to electrically couple the channel output circuit to the cryogenic DUT; at least one resistor positioned along a transmission path that couples a pattern generator circuit to the channel output circuit that includes the channel select circuit; a controller configured to: cause respective test current signals to be provided by the pattern generator circuit along the transmission path when the channel select circuit is in respective first and second states to establish respective first and second voltages across the at least one resistor; determine first and second impedance characteristics of the transmission path based on the established respective first and second voltages and an amount of current being provided by respective test current signals; determine an impedance of the cryogenic DUT based on the first and second determined impedance characteristics of the transmission path; receive a first determined average voltage based on measurements of the first voltage across the at least one resistor when the channel output circuit is in the first state; and receive a second determined average voltage based on measurements of the second voltage across the at least one resistor when the channel output circuit is in the second state.
2. The system of claim 1, wherein the transmission path has a first impedance when the channel select circuit is in the first state, and the transmission path has a second impedance when the channel select circuit is in the second state, wherein the cryogenic changes an impedance of the transmission path from the first impedance to the second impedance in response to being electrically coupled to the transmission path when the channel select circuit is in the second state.
3. The system of claim 1, further comprising a voltage sensing circuit to measure the first and second voltage across the at least one resistor and a voltage calculator circuit configured to determine the first determined average voltage based on measurements of the first voltage across the at least one resistor, and determine the second determined average voltage based on measurements of the second voltage across the at least one resistor.
4. The system of claim 3, wherein respective test current signals includes first and second current test signals, the controller being configured to determine the first impedance characteristic of the transmission path based on the first average voltage and an amount of current being provided by the first current test signal, and determine the second impedance characteristic of the transmission path based on the second average voltage and an amount of current being provided by the second current test signal.
5. The system of claim 4, wherein the controller is configured to subtract the second impedance characteristics of the transmission path from the first impedance characteristics of the transmission path to compute an impedance difference representative of the impedance of the cryogenic DUT.
6. The system of claim 5, further comprising a current sensing circuit configured to measure the amount of current being provided as the first current test signal and the second current test signal, respectively.
7. The system of claim 6, wherein the transmission path comprises a first transmission path that extends from a first output of the pattern generator circuit to a first input of the channel select circuit and a second transmission path that extends from a second output of the pattern generator circuit to a second input of the channel select circuit.
8. The system of claim 7, wherein the at least one resistor is positioned along a respective one of the first and second transmission paths, and the system further comprises another resistor positioned along one of the remaining first and second transmission paths.
9. The system of claim 8, wherein the channel select circuit comprises a relay having first and second inputs corresponding to the first and second inputs of the channel select circuit, the relay be configured to short the first and second inputs to form a closed loop circuit between the pattern

generator circuit and the channel select circuit in response to a channel control signal provided by the controller to set the channel select circuit to the first state.
10. A circuit comprising: a measuring circuit configured to: measure a first voltage established across a resistor along a transmission path based on a first current signal, wherein the first current test signal is provided along the transmission path when a channel output circuit is isolated from a cryogenic device under test (DUT), wherein the transmission path couples a pattern generator circuit for generating current signals to a channel output circuit; measure a second voltage established across the resistor based on a second current signal, wherein the second current test signal is provided along the transmission path when the channel output circuit is electrically coupled to cryogenic DUT; determine first and second average voltages based on the measurements of the first and second voltages, respectively; a controller configured to: receive the first and second determined average voltages; determine first and second impedance characteristics of the transmission path based on respective first and second determined average voltages and an amount of current being provided by respective first and second current test signals; and determine an impedance of the cryogenic DUT based on the first and second determined impedance characteristics of the transmission path.
11. The system of claim 10, wherein the transmission path has a first impedance when the channel output circuit is isolated from the cryogenic DUT, and the transmission path has a second impedance when the channel output circuit is electrically coupled to the cryogenic DUT, wherein the cryogenic DUT changes an impedance of the transmission path from the first impedance to the second impedance in response to being electrically coupled to the transmission path by the channel output circuit.
12. The system of claim 11, wherein the measuring circuit comprises a voltage sensing circuit to measure the first and second voltages across the resistor and a voltage calculator circuit configured to determine the first and second average voltages based on respective measurements of the first and second voltages.
13. The system of claim 12, wherein the controller is configured to subtract the second impedance characteristics of the transmission path from the first impedance characteristics of the transmission path to compute an impedance difference representative of the impedance of the cryogenic DUT.
14. The system of claim 13, wherein the transmission path comprises a first transmission path that extends from a first output of the pattern generator circuit to a first input of the channel select circuit and a second transmission path that extends from a second output of the pattern generator circuit to a second input of the channel select circuit.
15. The system of claim 14, wherein the resistor is positioned along a respective one of the first and second transmission paths, and the system further comprises another resistor positioned along one of the remaining first and second transmission paths.
16. The system of claim 15, wherein the channel select circuit comprises a relay having first and second inputs corresponding to the first and second inputs of the channel select circuit, the relay be configured to short the first and second inputs to form a closed loop circuit between the pattern generator circuit and the channel select circuit in response to a channel control signal provided by the controller.
17. A method comprising: causing a channel output circuit to be electrically isolated from a cryogenic device under test (DUT); controlling a test pattern generator circuit to provide a first current test signal along a transmission path so that the first current test signal flows through at least one resistor located in the transmission path to establish a first voltage across the at least one resistor; receiving a first determined average voltage based on measurements of the first voltage across the at least one resistor; determining first impedance characteristics of the transmission path based on the determined first average voltage and an amount of current being provided by the first current test signal; causing the channel output circuit to be electrically coupled to the cryogenic DUT such that the cryogenic DUT is part of the transmission path in response to determining the

first impedance characteristics of the transmission path; controlling the test pattern generator circuit to provide a second current test signal along the transmission path so that the second current test signal flows through the cryogenic DUT and the at least one resistor to establish a first voltage across the at least one resistor; receiving a second determined average voltage based on measurements of the second voltage across the at least one resistor; determining second impedance characteristics of the transmission path based on the second average voltage and an amount of current being provided by the second current test signal; and determining impedance characteristics of the cryogenic DUT based on the first and second impedance characteristic of the transmission path.

18. The method of claim 17, wherein causing the channel output circuit to be electrically coupled to the cryogenic DUT such that the cryogenic DUT is part of the transmission path comprises providing a channel control signal for a channel select circuit of the channel select circuit to cause the channel select circuit to switch from a first state to a second state, wherein in the first state the channel output circuit is electrically isolated from the cryogenic DUT and in the second state channel output circuit is electrically coupled to the cryogenic DUT.

19. The method of claim 18, wherein the determining impedance characteristics of the cryogenic DUT based on the first and second impedance characteristic of the transmission path comprises subtracting the second impedance characteristics of the transmission path from the first impedance characteristics of the transmission path to compute an impedance difference representative of the impedance of the cryogenic DUT.